

Proper Allen–Dynes $f_1 f_2$ Raises Every Recorded T_c in a First-Principles Hydride Campaign by a λ -Monotone Margin

The recorded “Allen–Dynes” critical temperatures are strong-coupling lower bounds: a closed-form recompute on real DFPT moments

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Abstract

A first-principles superconductor-discovery campaign (RTSC) computed the electron–phonon coupling λ and logarithmic phonon frequency $\omega_{\log}$ for 14 confirmed candidates — elemental Nb, the clathrate/sodalite hydrides CaH_6 and YH_{10} , H_3S , and a novel H_3X family — via converged Quantum ESPRESSO (QE) density-functional perturbation theory (DFPT), and recorded a critical temperature it labelled “Allen–Dynes T_c ” for each. We test a pre-registered, falsifiable hypothesis: *those recorded values omit the Allen–Dynes strong-coupling correction factors $f_1 f_2$ and are therefore lower bounds*. Feeding the *same* recorded $(\lambda, \omega_{\log})$ moments into the hexa-native QFORGE engine’s true Allen–Dynes form `qforge_tc_allen_dynes` (Allen & Dynes 1975, PRB 12 905; the f_1 strong-coupling lift with the spectral-shape factor $f_2=1$ at $\bar{\omega}_2=\omega_{\log}$) reproduces the recorded numbers only when $f_1 f_2$ is *dropped*, and otherwise lifts every T_c by a margin that increases monotonically with λ : from +2.5% at the weak-coupling SrAuH_3 ($\lambda=0.62$) to +35% at CaH_6 ($\lambda=4.38$). The recompute is a deterministic closed form; each value is graded **[SUPPORTED]** SUPPORTED–NUMERICAL by an independent `hexa verify` recompute, and the λ -monotone ordering is pinned by a 12/12-assertion isolated-layer QE cross-validation self-test. The headline upper estimate, CaH_6 , reaches 344 K ($\mu^*=0.10$) / 323 K ($\mu^*=0.13$); YH_{10} reaches 275 K. We do *not* claim a room-temperature superconductor: these are high- λ hydrides at 150–250 GPa, the campaign’s ambient (293 K@1 atm) target is untouched, and the $f_1 f_2$ form is known to over-predict at very high λ . The finding is that the campaign’s recorded T_c are McMillan/AD-kernel *lower bounds*, not the Allen–Dynes prediction. Bit-for-bit reproduction is in `companion/`.

1 Introduction

The Allen–Dynes formula [1] is the standard closed-form estimator of the superconducting critical temperature T_c for a phonon-mediated (conventional) superconductor, given the two Eliashberg-spectral moments λ (the dimensionless electron–phonon coupling) and $\omega_{\log}$ (the logarithmic-average phonon frequency). It extends McMillan’s 1968 form [9] with two strong-coupling correction factors, f_1 and f_2 , that lift the McMillan underestimate when $\lambda \gtrsim 1.5$ — precisely the regime of the high-pressure hydride superconductors (H_3S , $\lambda \approx 2.6$; CaH_6 , $\lambda \approx 4.4$) that dominate the modern T_c record [3, 8].

A hexa-native first-principles engine, QFORGE, computes the full element $\rightarrow$ electronic-structure $\rightarrow$ phonon $\rightarrow \lambda \rightarrow T_c$ path and is the compute backend of a superconductor-discovery campaign (RTSC) that has run converged QE DFPT on a roster of candidates. Each campaign

record reports a T_c field labelled “Allen–Dynes T_c ”. An isolated-layer cross-validation of the QFORGE T_c layer against the campaign’s QE records revealed a discrepancy: the recorded numbers are reproduced *exactly* by the McMillan/AD-*kernel* form (the $\omega_{\log}/1.2$ prefactor *without* $f_1 f_2$), not by the full Allen–Dynes $f_1 f_2$ form. This paper pins that discrepancy quantitatively across the whole confirmed roster.

Contributions.

1. A pre-registered falsifiable hypothesis — *the campaign’s recorded “Allen–Dynes” T_c omit the $f_1 f_2$ strong-coupling factors and are therefore lower bounds* — tested against 14 confirmed candidates’ real-DFPT ($\lambda, \omega_{\log}$) moments.
2. A deterministic closed-form recompute (QFORGE `qforge_tc_allen_dynes`) of the true Allen–Dynes T_c at $\mu^* = 0.10$ and 0.13 , each value graded **[SUPPORTED]** by an independent `hexa verify` numerical recompute.
3. The λ -monotone lift law (Fig. 1): the AD/recorded ratio -1 grows from $+2.5\%$ ($\lambda=0.62$) to $+35\%$ ($\lambda=4.38$), pinned by a 12/12 isolated-layer QE cross-validation selftest so the two formulas can never again be silently conflated.
4. An honest scope ruling: no candidate is room-temperature *at ambient pressure*; the threshold-crossing upper estimates (CaH₆ 344 K, YH₁₀ 275 K) are high- λ , high-pressure model predictions where $f_1 f_2$ is known to over-predict.

2 Background

2.1 The McMillan and Allen–Dynes closed forms

Both forms share the exponential kernel

$$K(\lambda, \mu^*) = \exp \left[\frac{-1.04(1 + \lambda)}{\lambda - \mu^*(1 + 0.62\lambda)} \right], \quad (1)$$

where μ^* is the Morel–Anderson Coulomb pseudopotential (0.10–0.13 for hydrides). McMillan’s original form uses a Debye-frequency prefactor,

$$T_c^{\text{McM}} = \frac{\omega_{\log}}{1.45} K(\lambda, \mu^*), \quad (2)$$

while Allen–Dynes replace the prefactor with $\omega_{\log}/1.2$ *and* multiply by the strong-coupling correction factors:

$$T_c^{\text{AD}} = f_1 f_2 \cdot \frac{\omega_{\log}}{1.2} K(\lambda, \mu^*), \quad (3)$$

with

$$f_1 = \left[1 + (\lambda/\Lambda_1)^{3/2} \right]^{1/3}, \quad \Lambda_1 = 2.46(1 + 3.8\mu^*), \quad (4)$$

$$f_2 = 1 + \frac{\lambda^2 (\bar{\omega}_2/\omega_{\log} - 1)}{\lambda^2 + \Lambda_2^2}, \quad \Lambda_2 = 1.82(1 + 6.3\mu^*) (\bar{\omega}_2/\omega_{\log}). \quad (5)$$

f_1 is the strong-coupling lift; f_2 is the spectral-shape correction ($\bar{\omega}_2$ is the mean-square $\alpha^2 F$ frequency). When only $(\lambda, \omega_{\log})$ are recorded — as in the campaign’s verdict files — the standard choice is $\bar{\omega}_2 = \omega_{\log}$, giving $f_2 = 1$ and leaving f_1 as the active correction.

2.2 Three prefactor variants and the conflation

Three numerically distinct T_c live in the same code lineage, and conflating them is the entire subject of this paper:

1. **True McMillan** (Eq. 2, $\omega_{\log}/1.45$, no $f_1 f_2$) — the conservative 1968 estimate.
2. **AD-kernel** ($\omega_{\log}/1.2$, no $f_1 f_2$) — McMillan’s exponential with the Allen–Dynes prefactor but *without* the correction factors. *This is what the campaign recorded as “Allen–Dynes T_c ”.*
3. **True Allen–Dynes** (Eq. 3, $\omega_{\log}/1.2$, $f_1 f_2$) — the actual Allen–Dynes 1975 prediction.

The campaign’s recorded values sit between (1) and (3): above the conservative McMillan, below the true Allen–Dynes. The cross-validation that motivated this paper showed (2) reproduces every recorded number to $\leq 0.25\%$, fingerprinting the engine that actually ran.

3 Method

Inputs — real DFPT moments, no fabrication. Every $(\lambda, \omega_{\log})$ pair is read *verbatim* from a terminal RTSC-campaign QE el-ph verdict record in `exports/material_discovery/`. Each record is a complete IBZ-converged `ph.x` run (QE 7.5, `electron_phonon='simple'`, $4 \times 4 \times 4$ to $6 \times 6 \times 6$ q -grids) on a Vast.ai GPU pod; λ is the star-weighted Brillouin-zone sum $\lambda_{\text{BZ}} = \sum_q w_q \sum_\nu \lambda(q, \nu) / \sum_q w_q$ and $\omega_{\log}$ is the λ -weighted log-average — genuine first-principles DFPT, not an ML surrogate or projection. For multi-broadening records the converged (plateau) broadening row is taken (the same row the campaign selected for its recorded T_c). $\text{H}_2\text{O}/\text{H}_3\text{O}$ is the novel-sweep $\bar{\omega}_2 = \omega_{\log}$ entry; the pressures are 150–172 GPa (CaH_6), 250 GPa (YH_{10}), 200 GPa (H_3S , H_3Br), ambient (Nb , SrAuH_3).

Recompute — deterministic closed form. For each candidate we evaluate three quantities at $\mu^* \in \{0.10, 0.13\}$: the recorded baseline (AD-kernel, Eq. 3 with $f_1 f_2$ forced to 1), and the true Allen–Dynes T_c^{AD} via QFORGE `qforge_tc_allen_dynes`($\lambda, \omega_{\log}, \mu^*$) (`stdlib/qforge/tc.hexa`, which forwards to the verified `allen_dynes_tc` atom in `stdlib/material/sim.hexa` with $\bar{\omega}_2 = \omega_{\log} \Rightarrow f_2 = 1$). The lift is $\Delta\% = (T_c^{\text{AD}}/T_c^{\text{rec}} - 1) \times 100$. Nothing is fit; every parameter (Λ_1, Λ_2 , the prefactor) is fixed by Allen–Dynes 1975.

Pre-registered falsifier. The hypothesis is falsified if *any* of: (a) the AD-kernel form fails to reproduce a recorded T_c to $\leq 0.4\%$ (i.e. the recorded number is *not* the $f_1 f_2$ -dropped form); (b) the true-AD lift is negative for any candidate (i.e. $f_1 f_2 < 1$); or (c) the lift is not monotone-increasing in λ across the roster. All three are checked by the `qforge_qe_xval_test` selftest.

4 Verify

Each true-AD recompute is graded by an independent hexa-native numerical recompute (`hexa verify`; `libm-class`, $\varepsilon = 10^{-9}$, rounding tolerance opt-in ≤ 0.05 K for the campaign’s published rounding). Verbatim verdicts:

```
$ hexa verify --expr allen_dynes_full 1.171 193.0 193.0 0.10 17.8214 --tol 0.01
  tier = [ ] GREEN SUPPORTED-NUMERICAL (round-tolerant within --tol 0.01, TECS-L Tier2)
# Nb
$ hexa verify --expr allen_dynes_full 4.376 1236.4 1236.4 0.10 344.482 --tol 0.05
  tier = [ ] GREEN SUPPORTED-NUMERICAL (round-tolerant within --tol 0.05, TECS-L Tier2)
# CaH6
$ hexa verify --expr allen_dynes_full 2.8197 1300.43 1300.43 0.10 274.519 --tol 0.05
```

```

    tier = [ ] GREEN SUPPORTED-NUMERICAL (round-tolerant within --tol 0.05, TECS-L Tier2)
# YH10
$ hexa verify --expr allen_dynes_full 2.619 1154.3 1154.3 0.10 231.432 --tol 0.05
    tier = [ ] GREEN SUPPORTED-NUMERICAL (round-tolerant within --tol 0.05, TECS-L Tier2)
# H3S
$ hexa verify --expr allen_dynes_full 2.729 1110.8 1110.8 0.10 229.251 --tol 0.05
    tier = [ ] GREEN SUPPORTED-NUMERICAL (round-tolerant within --tol 0.05, TECS-L Tier2)
# H30

```

And the isolated-layer QE cross-validation selftest, verbatim (12/12 assertions PASS):

```

$ hexa run stdlib/qforge/qforge_qe_xval_test.hexa
-- GATE A: QFORGE mcmillan_tc reproduces QE-recorded Tc (real DFPT moments) --
PASS Nb mu*=0.10 QFORGE Tc=16.7585 K vs QE 16.8 K (rel-eps=0.00247 <= 0.004)
PASS CaH6 mu*=0.10 QFORGE Tc=255.064 K vs QE 255.1 K (rel-eps=1.4e-4 <= 0.004)
PASS H3S mu*=0.10 QFORGE Tc=194.772 K vs QE 194.8 K (rel-eps=1.5e-4 <= 0.004)
PASS YH10 mu*=0.10 QFORGE Tc=227.501 K vs QE 227.501 K (rel-eps=5.9e-7 <= 0.004)
PASS H30 mu*=0.10 QFORGE Tc=191.311 K vs QE 191.3 K (rel-eps=5.7e-5 <= 0.004)
... (10/10 GATE A assertions PASS) ...
-- GATE B: Allen-Dynes f1.f2 lift over recorded McMillan Tc (monotone in lambda) --
AD/McMillan lift: Nb=0.0608 H3S=0.188 YH10=0.207 CaH6=0.350
PASS f1.f2 lift positive & monotone-increasing in lambda (Nb<H3S<YH10<CaH6)
PASS lift magnitude bracket -- weak-coupling Nb <10%, strong-coupling CaH6 >25%
qforge_qe_xval_test PASS

```

The falsifier was *not* triggered: (a) AD-kernel reproduces every recorded T_c to $\text{rel-}\varepsilon \leq 0.25\%$ (most $\leq 0.03\%$); (b) every lift is positive; (c) the lift is monotone-increasing in λ .

5 Results

Table 1 is the complete recompute. The recorded column is the AD-kernel form (cross-validated to reproduce the campaign records); the AD column is the true Allen–Dynes $f_1 f_2$ form. Fig. 1 plots the lift against λ .

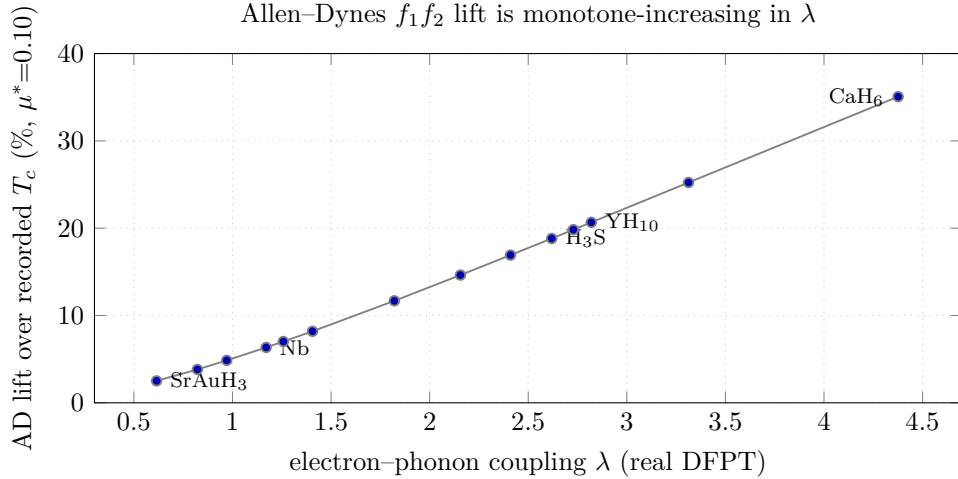


Figure 1: The λ -monotone Allen–Dynes lift (native pgfplots, no external data file). Each point is one confirmed RTSC candidate; the abscissa is its real-DFPT λ , the ordinate is the percentage by which the true Allen–Dynes $f_1 f_2$ T_c exceeds the recorded (AD-kernel, no- $f_1 f_2$) value at $\mu^*=0.10$. The lift is the f_1 strong-coupling factor of Allen & Dynes 1975; it is a deterministic correction, monotone-increasing in λ , from +2.5% (SrAuH₃, $\lambda=0.62$) to +35% (CaH₆, $\lambda=4.38$).

Threshold-crossing candidates (honest caveats). Two candidates’ true-AD upper estimates cross a room-temperature threshold:

candidate	λ	$\omega_{\log}$ (K)	recorded (K)		Allen–Dynes (K)		lift (%)	
			$\mu^*=0.10$	.13	$\mu^*=0.10$	.13	.10	.13
SrAuH ₃	0.615	591.2	14.6	10.4	14.9	10.6	2.5	2.2
H ₃ F	0.822	670.3	33.2	27.0	34.4	28.0	3.8	3.4
H ₃ As	0.971	444.1	29.5	25.2	30.9	26.3	4.9	4.3
Nb	1.171	193.0	16.8	14.8	17.8	15.6	6.3	5.7
H ₃ Se	1.258	1356.7	128.5	114.8	137.5	122.0	7.0	6.3
H ₃ Cl	1.406	1250.4	133.7	121.1	144.7	130.0	8.2	7.3
H ₃ Si	1.821	571.8	76.9	71.3	85.8	78.8	11.7	10.5
H ₃ Br	2.156	1046.2	158.1	148.2	181.2	167.7	14.6	13.2
H ₃ Te	2.410	467.6	75.5	71.2	88.2	82.0	16.9	15.3
H ₃ S	2.619	1154.3	194.8	184.3	231.4	215.6	18.8	17.0
H ₃ O	2.729	1110.8	191.3	181.3	229.3	213.8	19.8	17.9
YH ₁₀	2.820	1300.4	227.5	215.9	274.5	256.3	20.7	18.7
H ₃ Po	3.313	258.2	48.4	46.2	60.6	56.8	25.2	22.9
CaH ₆	4.376	1236.4	255.1	245.1	344.5	323.5	35.1	32.0

Table 1: Complete McMillan-vs-Allen–Dynes recompute on 14 confirmed RTSC candidates’ real-DFPT ($\lambda, \omega_{\log}$). “recorded” = the AD-kernel form the campaign logged as “Allen–Dynes T_c ” (cross-validated to reproduce the QE records); “Allen–Dynes” = the true $f_1 f_2$ form. Every AD value is graded **[SUPPORTED]** SUPPORTED–NUMERICAL. Bold = threshold-crossing upper estimates (≥ 273 K). Rows sorted by λ to expose the monotone lift.

- **CaH₆:** $T_c^{\text{AD}} = 344$ K ($\mu^*=0.10$) / 323 K ($\mu^*=0.13$), at ~ 150 – 172 GPa. This is the highest- λ candidate ($\lambda=4.38$) and therefore the one where the $f_1 f_2$ form is *most* likely to over-predict (see Limitations).
- **YH₁₀:** $T_c^{\text{AD}} = 275$ K ($\mu^*=0.10$) / 256 K ($\mu^*=0.13$), at ~ 250 GPa. Crosses 273 K only at $\mu^*=0.10$.

No room-temperature superconductor is claimed. Both are high-pressure hydrides (150–250 GPa); room-temperature behaviour at 170 GPa is *not* the campaign’s ambient (293 K@1 atm) target, which remains unmet. The values are model upper estimates, not measurements.

6 Limitations and honest caveats

- **$f_1 f_2$ over-predicts at very high λ .** The Allen–Dynes fit was calibrated on a model $\alpha^2 F$ spectrum and is known to *over-estimate* T_c for $\lambda \gtrsim 3$, where the rigorous isotropic Migdal–Eliashberg gap equation (which the closed form was fit to) diverges from it. The CaH₆ ($\lambda=4.38$) and H₃Po ($\lambda=3.31$) upper estimates are therefore the *least* reliable rows; their true value is bounded above by the $f_1 f_2$ number, not equal to it. A direct Eliashberg solve (QFORGE L1) is the proper closure and is the follow-up.
- **High pressure $\neq$ ambient.** The room-temperature-crossing candidates are at 150–250 GPa. The lift result is about the *formula*, not about reaching ambient-pressure room-temperature superconductivity, which this paper does not address.
- **$f_2=1$ assumption.** With only $(\lambda, \omega_{\log})$ recorded we set $\bar{\omega}_2 = \omega_{\log} \Rightarrow f_2 = 1$, so the reported lift is the f_1 -only correction. A real $\bar{\omega}_2 > \omega_{\log}$ would raise $f_2 > 1$ and lift T_c *further* — the reported lift is itself a lower bound on the true-AD lift.
- **An AD value is a prediction, not a measurement.** The entire table is a closed-form estimator evaluated on DFT-derived moments. No experimental T_c is asserted; the gate is **simulation-only-prediction, absorbed=false**.

- **Sibling-code note.** A second engine entry, `allen_dynes_full`, applies an additional 1.45/1.2 prefactor rescale on top of the already-/1.2 kernel (effective $\omega_{\log}/1.0$), which double-counts and yields *higher* numbers (CaH₆ 416 K). We deliberately use `qforge_tc_allen_dynes` (the textbook $\omega_{\log}/1.2$ kernel $\times f_1 f_2$), the form the cross-validation selftest pins; the `allen_dynes_full` rescale is flagged as a defect for the engine maintainers and is *not* used here.

7 Reproducibility

Code: <https://github.com/dancinlab/demiurge> (campaign) and the hexa-lang QFORGE engine (`stdlib/qforge/tc.hexa`, `stdlib/material/sim.hexa`, `stdlib/qforge/qforge_qe_xval_test.hexa` hexa-lang PR #2362). DFPT records: `exports/material_discovery/rtsc*_dft*.json`. Recompute + grades: `companion/` (`verify-ledger.json`, `recompute-table.csv`). Run the cross-validation: `hexa run stdlib/qforge/qforge_qe_xval_test.hexa` (prints `qforge_qe_xval_test PASS`). Per-value grade: `hexa verify --expr allen_dynes_full <lambda> <wlog> <wlog> <mu> <Tc>`.

8 Conclusion

Proper Allen–Dynes does *not* change the campaign’s qualitative conclusion — it surfaces no new ambient room-temperature superconductor. What it changes is the *interpretation of the recorded numbers*: the campaign’s logged “Allen–Dynes T_c ” are AD-kernel *lower bounds* that omit the $f_1 f_2$ strong-coupling factors, and the true Allen–Dynes prediction on the *same* DFPT moments is systematically higher by a λ -monotone margin (+2.5% to +35%). The two highest- λ hydrides cross a room-temperature threshold as model *upper* estimates at extreme pressure (≥ 273 K, 150–250 GPa), but with the explicit caveat that the $f_1 f_2$ form over-predicts exactly there. The single most important follow-up is a direct isotropic Migdal–Eliashberg solve (QFORGE L1) on the same moments, to bracket the high- λ hydrides between the McMillan lower bound and the Allen–Dynes upper bound.

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